

A380X MFD ATC COM pages and the SD mailbox

Every ATC COM page of the A380 FCOM on the FBW ATSU, and the SD mailbox to answer the ATC

Branch	pr/22-a380x-mfd-atccom
Aircraft	A380X (MFD, SD)
Commits	b3a72399e feat(a380x/mfd): ATC COM pages and the SD mailbox per the A380 FCOM; c66b105f fix(a380x/mfd): lint and strict types of the ATC COM pages and the SD mailbox; 703024a90 docs(changelog): add the ATC COM pages and the SD mailbox; fixes of the in-sim test: 899b8c6eb frame delete, label layout, ATIS auto update; c0326bbd0 texts sized for the FBW display font; 625a792d5 ATIS time from the ATIS message; 2026-09-29: 40b6f21d3 fix(atsu): SEND FAILED when no ATC center is active; 591aa0b5d fix(a380x/fms): connect the datalink networks with the active flight number
Base	pr/10-a380x-mfd-fcom-pages (MFD FCOM layout and common components)
Status	CI checks pass. Tested in the sim: ATIS pages and the REQUEST / REPORT / EMERGENCY frames (2026-09-27); logon to BeyondATC (RJJJ) and a MAYDAY sent after the datalink connection fix (2026-09-29).

Purpose

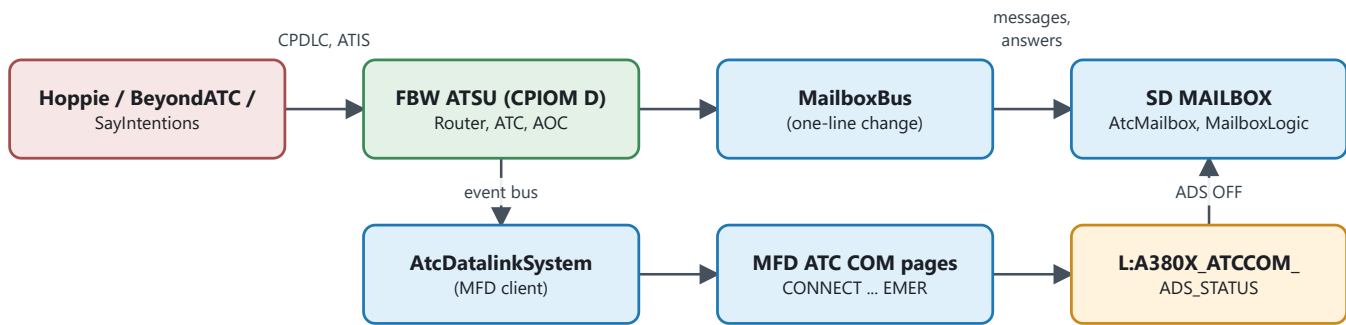
The A380X MFD had placeholder ATC COM pages (connect, D-ATIS, message record) and the SD mailbox could not answer the ATC. The FBW ATSU (Hoppie, BeyondATC, SayIntentions through the flyPad ATSU/AOC settings) was already hosted by the A380X systems host.

Specification (A380 FCOM DSC-46-10-20-30 ATC COM pages, DSC-46-10-10-70 mailbox, DSC-46-10-20-40 messages)

Page	Content
Header	CONNECT / REQUEST / REPORT & MODIFY / MSG RECORD / ATIS / EMER, the ATC COM system menu
CONNECT	NOTIFICATION (logon: NOTIFYING, NOTIFICATION FAILED, the centers notified), CONNECTION STATUS (with the ADS status), MAX UPLINK DELAY
ATIS	LIST with three request areas (departure, arrival, alternate, following the FMS airports), AUTO UPDATE, UPDATE ALL, ATIS RECEIVED
MSG RECORD	LIST and ZOOM, ERASE ALL of the closed dialogues, MSG RECORD USED OFFSIDE
REQUEST	every FCOM frame mapped to its CPDLC downlink element, the DEPARTURE (DCL) and OCEANIC (OCL) clearances, XFR TO MAILBOX
REPORT & MODIFY	AUTO and MANUAL POSITION reports filled by the FMS, OTHER REPORTS, MODIFY of a prepared reply
EMERGENCY	MAYDAY, PANPAN, CANCEL EMER with the emergency frames; MAYDAY switches the ADS to EMERGENCY
SD MAILBOX	answer buttons per expected response (WILCO / STANDBY / UNABLE, AFFIRM / NEGATIVE, ROGER), SEND / CANCEL / CLOSE, DUE TO and FREETEXT replies, ACK / REFUSE for DCL and OCL, the FCOM status colours

 Component added or changed by the feature  MSFS / simulator  Aircraft system (systems host, FMS, FWS)  Display / page

 Data, LVar, storage  Crew, flypad, external program



Architecture

Interfaces

LVar	Written by	Meaning
L:A380X_ATCCOM_ADS_STATUS	MFD	0 ARMED, 1 CONNECTED, 2 OFF; displayed by the SD mailbox
L:A380X_MFD_{side}_ATCCOM_MSG_RECORD_ZOOM	MFD	an MFD shows a recorded message: MSG RECORD USED OFFSIDE on the other one

Documented in `fbw-a380x/docs/a380x-private-local-vars.md` (46 - Information Systems). `fbw-common` MailboxBus: `downlinkTransmit` on an uplink sends its existing open or failed response.

Tests

33 unit tests: request frames, entry formats, ATIS text and time, message record, reports, mailbox logic.

Limitations

- No ADS contract through the ATSU: the ADS status is local (ARMED / OFF, EMERGENCY).
- Parked: LOAD to SEC3 of a route clearance (the ATSU has no route loading) and the KCCU MAILBOX key.
- The PRINT functions show PRINTER NOT AVAIL in this branch; the ATIS printing is on the feature branch and needs the pedestal printer, which has no PR yet.

Datalink connection and SEND FAILED (2026-09-29)

Item	Behaviour
FLT NBR (INIT)	the field updates its value at once (it wrote the flight plan only and refreshed on the next page)
Datalink networks	the FMS (FmsDatalinkConnection, FMC-A) connects TELEX and the ACARS provider whenever the flight number of the active flight plan changes, whatever its source; resent after 10 s without an answer of the router
FMS DATALINK NOT AVAIL	type II FMS message (DSC-22-FMS-20-110), white: connection refused, or no answer after two tries; cleared when connected or with no flight number; FLT NBR entered again retries
SEND FAILED	a downlink without an active ATC center shows SEND FAILED in the mailbox (DSC-46-10-20-60) instead of being dropped silently

PR description (from the PR draft)

From `pr-drafts/22-a380x-mfd-atccom.md`, section "Summary of Changes" onwards.

Summary of Changes

Every ATC COM page of the A380 FCOM (DSC-46-10-20-30) is laid out on its FCOM figure and wired to the FBW ATSU, and the SD MAILBOX (DSC-46-10-10-70) becomes a client of the ATSU mailbox protocol (the same one as the A32NX DCDU).

- **Header:** CONNECT / REQUEST / REPORT & MODIFY / MSG RECORD / ATIS / EMER, and the ATC COM system menu.
- **CONNECT:** NOTIFICATION (logon; NOTIFYING, NOTIFICATION FAILED; the centers notified), CONNECTION STATUS (with the ADS status), MAX UPLINK DELAY.
- **ATIS:** LIST, with three request areas (departure, arrival, alternate, following the FMS airports), AUTO UPDATE and UPDATE ALL, and ATIS RECEIVED.
- **MSG RECORD:** LIST and ZOOM, ERASE ALL of the closed dialogues, MSG RECORD USED OFFSIDE.
- **REQUEST:** every FCOM frame mapped to its CPDLC downlink element (climb and descent, altitude and block, direct, offset, weather deviation, heading and track, SID/STAR, speed, when can we expect, free text, voice contact, own separation, VMC descent, additional text), the DEPARTURE (DCL) and OCEANIC (OCL) clearances, XFR TO MAILBOX.
- **REPORT & MODIFY:** AUTO and MANUAL POSITION reports filled by the FMS, OTHER REPORTS, MODIFY of a prepared reply.
- **EMERGENCY:** MAYDAY, PANPAN and CANCEL EMER with the emergency frames; MAYDAY switches the ADS to EMERGENCY.
- **SD MAILBOX:** the answer buttons per expected response (WILCO / STANDBY / UNABLE, AFFIRM / NEGATIVE, ROGER), SEND / CANCEL / CLOSE, DUE TO and FREETEXT replies, ACK / REFUSE for DCL and OCL, the message status colours of the FCOM.
- FCOM entry formats and messages for every field.
- `fbw-common` MailboxBus (one line): `downlinkTransmit` on an uplink sends its existing open or failed response.
- The old placeholder ATC COM pages (connect, D-ATIS, message record) are removed.
- 33 unit tests (request frames, entry formats, ATIS text and time, message record, reports, mailbox logic).
- **Datalink connection (FMS):** the INIT page FLT NBR field wrote the flight plan only: the field did not refresh and a typed flight number never connected the datalink networks (TELEX and the ACARS provider), so every logon and downlink failed. The field now updates its value, and the FMS (`FmsDataLinkConnection`) connects the networks whenever the flight number of the active flight plan changes, whatever its source (INIT entry, company flight plan, SEC activation, flight reload), resending when the router does not answer. A refused connection, or no answer after two tries, shows the FMS message FMS DATALINK NOT AVAIL (DSC-22-FMS-20-110); entering the flight number again retries. 5 unit tests.
- **SEND FAILED:** a message sent from the mailbox without an active ATC center was dropped silently; the mailbox now shows SEND FAILED (DSC-46-10-20-60), in the shared ATC code (the A32NX DCDU too).

Cockpit API Changes

New private local vars, documented in `fbw-a380x/docs/a380x-private-local-vars.md` (46 - Information Systems):

- `L:A380X_ATCCOM_ADS_STATUS` (0 ARMED, 1 CONNECTED, 2 OFF), written by the MFD, displayed by the SD mailbox.
- `L:A380X_MFD_{side}_ATCCOM_MSG_RECORD_ZOOM` (an MFD shows a recorded message: MSG RECORD USED OFFSIDE on the other one).

Screenshots (if necessary)

TO ADD: each ATC COM page next to its FCOM figure (CONNECT pages, ATIS LIST, MSG RECORD LIST and ZOOM, REQUEST with frames, REPORT, EMERGENCY); the SD mailbox with an uplink and its answer buttons.

References

- A380 FCOM DSC-46-10-20-30 (ATC COM pages), DSC-46-10-10-70 (mailbox), DSC-46-10-20-40 (ATC COM messages); the CPDLC message elements (DM / UM numbers) of the FBW ATSU.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. flyPad: set the Hoppie ACARS network and logon code. MFD ATC COM > CONNECT > NOTIFICATION: notify a center: NOTIFIED TO CENTERS, then CONNECTION STATUS.
2. ATIS: request the departure ATIS; RECEIVED shows it; AUTO UPDATE.
3. REQUEST: add CLB TO and DIRECT frames, XFR TO MAILBOX; send from the SD mailbox; the reply buttons follow the uplink.
4. MSG RECORD: the dialogue appears; ZOOM; ERASE ALL keeps the open ones.
5. EMER: MAYDAY frame, XFR: ADS EMERGENCY.
6. `npm vitest run` on the six ATC COM / mailbox spec files (33 tests).